Removing a letter from the protein alphabet

Toward life with a 19–amino acid alphabet through generative artificial intelligence design

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Can essential cellular machinery work without isoleucine?

All living cells use at least 20 canonical amino acids. Several have similar chemistry, but apparent redundancy does not establish interchangeability.

1. Choose an amino acid that may be replaceable.
2. Redesign essential proteins while excluding that amino acid.
3. Test whether individually functional designs still work together.

The experiment changes protein sequences, rather than merely replacing synonymous codons.

Liu et al., Introduction; Fig. 1A–B.

The achievement has three distinct scales

Scale	What was demonstrated
Individual proteins	Ile removed from each of the 50 Ile-containing essential ribosomal proteins; 382 Ile sites in total.
Combined genomic locus	A viable strain carries a 21-gene ribosomal region built from a 19-amino acid alphabet.
Whole cell	A fully Ile-free proteome remains a future goal. Many proteins outside that region still contain Ile.

Abstract; Results, Figs. 3 and 5. Two of the 52 essential ribosomal proteins already lack Ile.

Why start with isoleucine?

Evolutionary evidence

- Ile, Val and Gln have low global conservation in the authors' ortholog analysis.
- Ile-to-Val is a frequently tolerated replacement.
- ESM2 and MSA substitution matrices agree strongly ($r = 0.92$).

Biochemical plausibility

- Ile and Val are hydrophobic branched-chain amino acids.
- Val costs less to biosynthesize in the cited accounting.
- Partial Val incorporation at Ile sites can be tolerated in prior work.

These observations nominate a target; none proves global dispensability.

Fig. 1C–D; main-text candidate selection; supplementary Fig. S1 described in main text.

A simple replacement baseline works in fewer than half

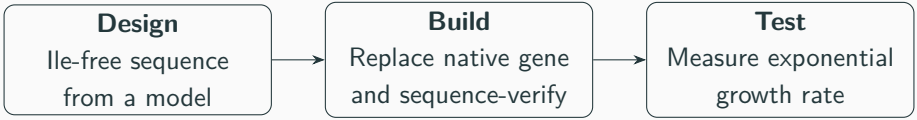
Initial experiment: replace every Ile with Val or Leu in 36 essential and 3 highly expressed genes, at their native loci.

- 17 of 39 genes tolerated complete replacement by at least one strategy.
- LpxD tolerated 36 Ile-to-Val changes; SecY tolerated 45 Ile-to-Leu changes.
- TrmD illustrates context dependence: Ile-to-Val preserved fitness, whereas Ile-to-Leu gave relative fitness 0.50.

Of the remaining 22 designs, 16 gave no colonies and 6 gave partial recombinants.

Fig. 1E; Results, "Testing systematic Ile replacement"; biological replicates $n = 6$.

Design–build–test makes function the final criterion



Relative fitness $= \mu_{\max,\mathrm{design}} / \mu_{\max,\mathrm{control}}$.

A single-protein design is called functionally recoded at fitness > 0.9. Controls account for the resistance cassette and, where applicable, synonymous recoding.

Materials and methods, genomic replacement and fitness measurements; Fig. 3.

Running example: mask Ile, then choose non-Ile residues

Native sequence	S T E L I V C R E K I A Q
Ile-only mask	S T E L ? V C R E K ? A Q
Illustrative output	S T E L V V C R E K L A Q

Simultaneous decoding: select all masked residues from the same context.

Sequential decoding: fill one site, update the sequence, then predict the next site.

All sampled sites exclude Ile. The displayed output is a teaching example, not an experimentally tested variant.

Fig. 2A–B; Materials and methods, sequence-based design.

Different models impose different constraints

Model	Information supplied	Design role
ESM2	Sequence context	Propose plausible replacements
MSA Transformer	Target plus homolog alignment	Use evolutionary covariation
ProteinMPNN	Backbone, including nearby chains	Generate sequences fitting structure
AfDesign	Target structural distances plus model bias	Optimize sequence through AlphaFold2

Sequence plausibility and structural compatibility provide complementary filters. Neither directly measures cell growth.

Fig. 2B–C; Materials and methods, structure processing and design protocols.

Selecting designs without reducing everything to one score

ProteinMPNN workflow

1. Generate 200 candidates per target across two temperatures and with/without sequence-neighbor redesign.
2. Score structural compatibility with AlphaFold2 and sequence plausibility with ESM2 pseudo-log-likelihood.
3. Test four candidates from the Pareto frontier: no alternative scores better on both criteria.

For a sequence x, $\mathrm{PLL}(x) = \sum_i \log P(x_i \mid x_{\setminus i})$.

PLL measures model preference; it is not a calibrated probability of viability.

Materials and methods, "Structure-aware design with ProteinMPNN selected based on Pareto front."

Compensation expands the search beyond Ile positions

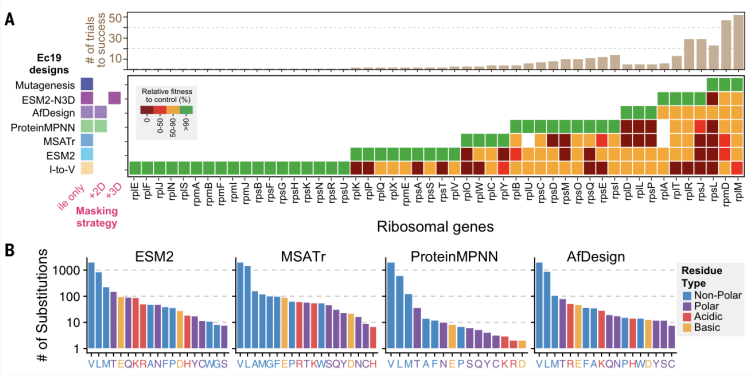
Replacing a buried side chain may require changing its neighbors.

- Sequence-neighbor designs allow local compensatory mutations.
- N3D masks additional positions within selected 3–7 Å distance windows around Ile in the experimental structure.
- A masked non-Ile position can retain its original residue.

Tradeoff: a wider search permits coordinated solutions, but excessive masking can produce implausible repetitive sequences. Candidates were filtered and ranked with AlphaFold2.

Fig. 2A; N3D Results and Materials and methods. Relevant supplementary panels were not directly accessible.

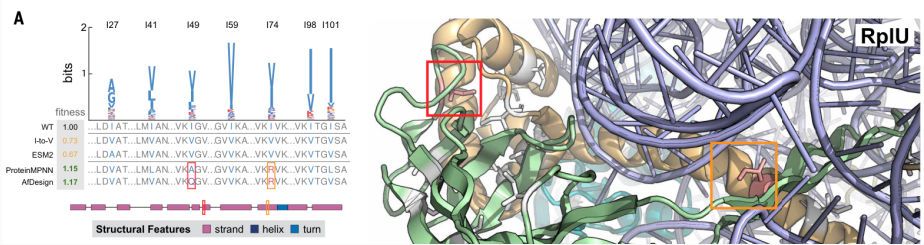
All 50 targets were solved, through several routes



Baseline solved 18 targets; ESM2 added 9, MSATr 5, ProteinMPNN 8, AfDesign 3, N3D 4, and mutagenesis 3.

Original Fig. 3A–B, cropped. Green: relative fitness $> 90\%$. Trial counts and the successive design strategy are shown.

Structural context can favor a surprising charged residue

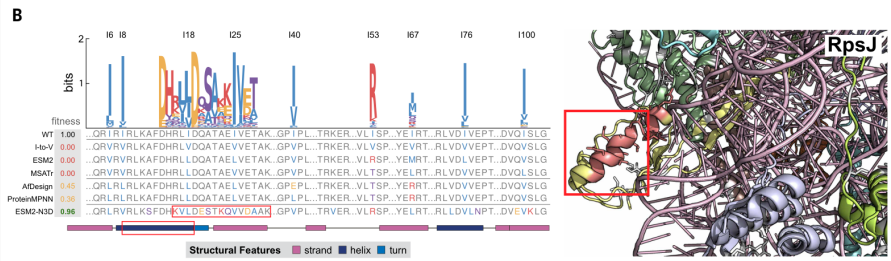


RplU I74R: predicted replacement probability was 0.001 from sequence alone, 0.2 with structure, and 0.4 with neighboring-chain context.

ProteinMPNN and AfDesign variants achieved relative fitness 1.15 and 1.17. The authors interpret this as structural context opening solutions rare in homologs; the mechanism is inferred rather than directly measured.

Original Fig. 4A, cropped; main-text interpretation; probabilities reported from Table S5 in main text.

RpsJ needs coordinated remodeling, not a single swap



ESM2-N3D redesigned a helix with eight non-Ile mutations alongside two Ile substitutions, reaching fitness 0.96.

The closest natural helix sequences still differed by at least five substitutions. This supports a new functional combination of evolutionary features, rather than direct copying of a natural Ile-free protein.

Original Fig. 4B, cropped; Fig. 4C–D and main-text sequence comparison.

AI proposals leave experimentally important blind spots

For *rpsL*, *rpmD* and *rplM*, testing 20–30 computational variants per gene did not recover near-control fitness.

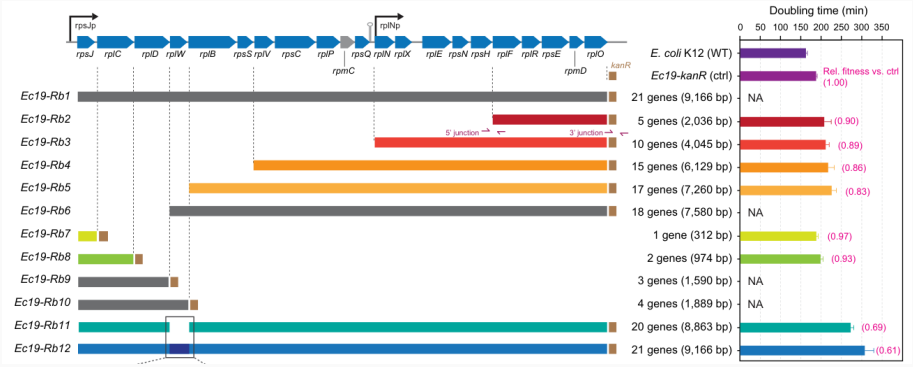
- The authors used NNS mutagenesis with MAGE to explore alternative Ile substitutions.
- Functional solutions included Gly, Gln, Trp and Phe replacements that models often disfavored.

The paper establishes a hybrid workflow, not a universally successful zero-shot predictor.

The sequential escalation between methods also prevents a clean, equally budgeted ranking of model performance.

Fig. 4E; main-text mutagenesis results. Last sentence is journal-club interpretation of the study design.

Combining designs reveals a second level of constraints



The first 21-gene cassette failed. Stepwise assembly localized a major obstacle to *rplW*; a 20-gene construct excluding it was viable at fitness 0.69.

Original Fig. 5C, cropped; M9 minimal medium, $n = 6$, error bars: standard error. NA: no viable construct recovered.

The failure point was also a DNA architecture problem

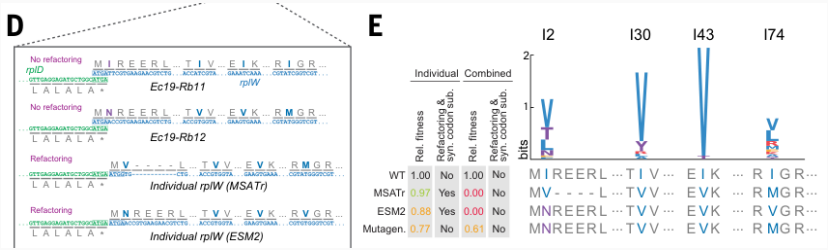
rplD and *rplW* share a 4-bp overlap involving the former's stop and the latter's second-position Ile codon.

1. Individual screening separated this overlap by sequence duplication.
2. Individually functional designs failed in the combined operon.
3. Preserving the native overlap restricted RplW position 2 to Thr or Asn.

Model consensus fixed I30V and I43V and suggested eight alternatives at I74: only $2 \times 1 \times 1 \times 8 = 16$ combinations to test. A protein-level design can fail because of its genomic context.

Fig. 5D–E; main-text failure analysis and model-guided mutagenesis.

A context-preserving variant rescues the full locus



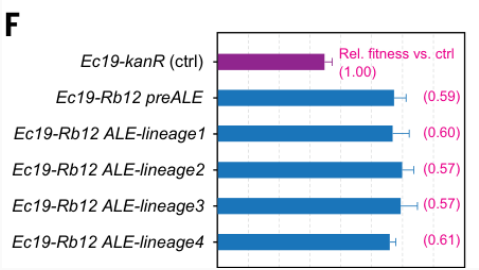
RplW Ec19_ID183: I2N, I30V, I43V, I74M.

- Relative fitness 0.77 individually with the native overlap; nonviable in the refactored screening context.
- Combined 21-gene strain Rb12: relative fitness 0.61 in M9 minimal medium.

This is substantial combined function with a substantial growth penalty.

Original Fig. 5D–E, cropped; Results. RpmC in the 21-gene locus was naturally Ile-free.

450 generations support stability under the tested conditions



Four independent Rb12 lineages evolved in LB in a turbidostat.

- No significant pre/post growth-rate difference ($P > 0.05$).
- Sequencing at 0, 150, 300 and 450 generations.
- 21–29 acquired mutations per lineage; none detected in the recoded locus and no Ile reversions.

Stability in this assay does not establish robustness across temperatures, nutrient limitations, stress or longer evolutionary histories.

Original Fig. 5F, cropped; ALE Results and Methods. Growth assay here is in LB, distinct from the M9 assembly comparison.

What the evidence supports, and what remains open

Supported by experiments	Still unresolved
Each essential ribosomal protein has a functional Ile-free solution.	Can all 50 redesigned proteins function simultaneously?
A 21-gene locus functions with a reduced amino acid alphabet.	Can >81,000 Ile sites across the proteome be removed?
Models reduce the experimental search burden.	Can models predict combined fitness and overlapping-gene constraints?
The combined locus persists over 450 generations in LB.	Can fitness and environmental robustness be restored?

Figs. 1D, 3–5; Discussion. Right column: journal-club questions.

Discussion: where should the next design effort go?

1. **Integration before scale?** Combine all ribosomal designs before attempting thousands of additional proteins.
2. **Protein versus DNA models?** Preserve translational coupling and model codon-level regulatory effects alongside structural compatibility.
3. **Prediction versus selection?** Use diverse models to narrow candidates, while retaining mutagenesis for model blind spots.

Take-home: a reduced amino acid alphabet is experimentally feasible for essential machinery, but protein design and genome design must be solved together.

Interpretation based on Liu et al., Discussion. Main full text read; supplementary Figs. S1–S17 and Tables S1–S8 could not be directly accessed.